FLOODWATCH

Reading Lagos Through Its Floods — A Data Story

Crowdsourced Flood Monitoring & Weather Severity Analytics for Lagos, Nigeria

Prepared by Chimbuchi Okey Ofoedu · Group 6 DA · July 2026

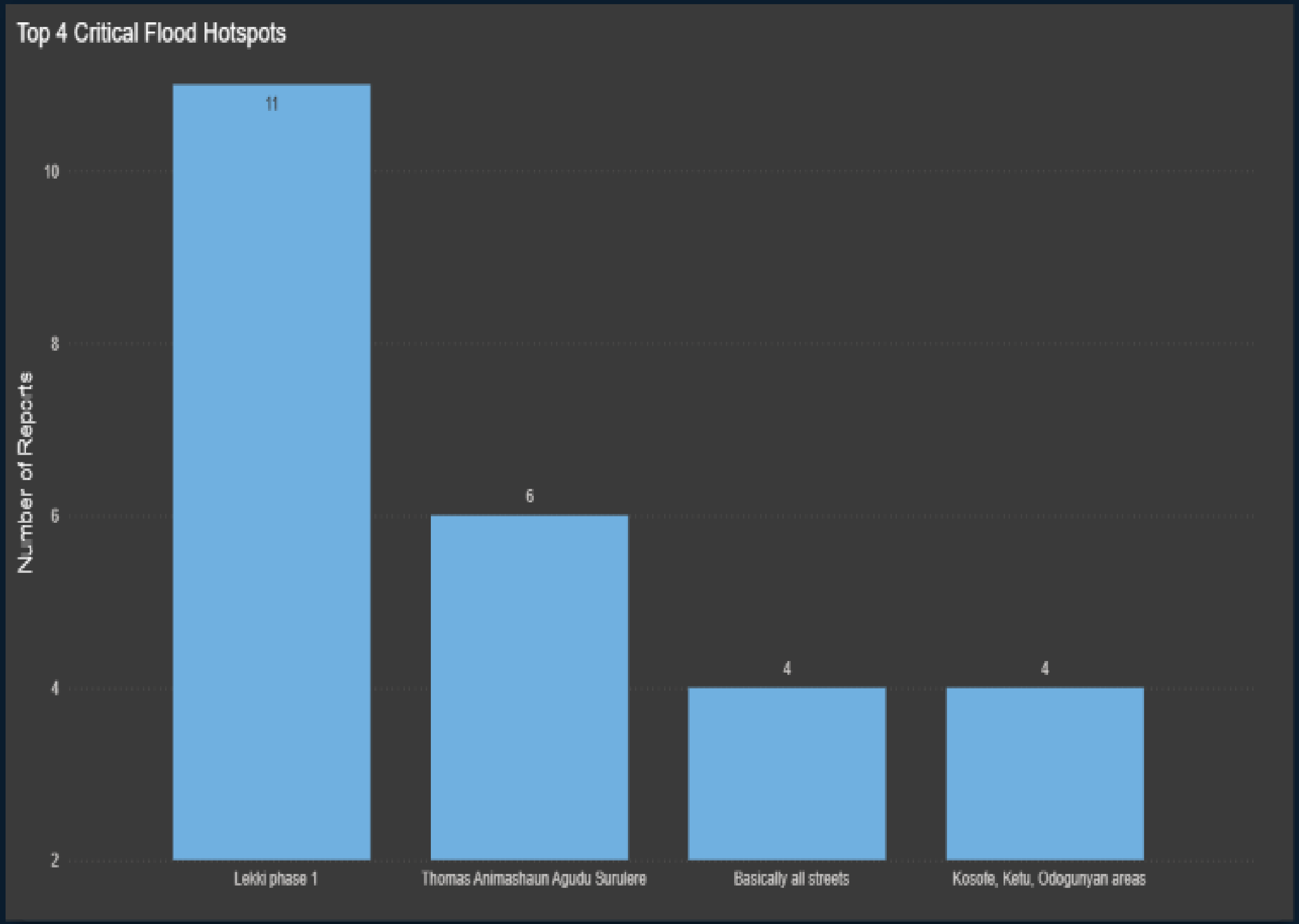
01 The Scale of the Problem

Over just a few weeks, Lagos residents flagged 78 flood incidents across the city. Eleven locations emerged as high-risk hotspots — many sitting at an average elevation of just 21 metres, low enough that a single heavy downpour can overwhelm them.

02

Where It Hits Hardest

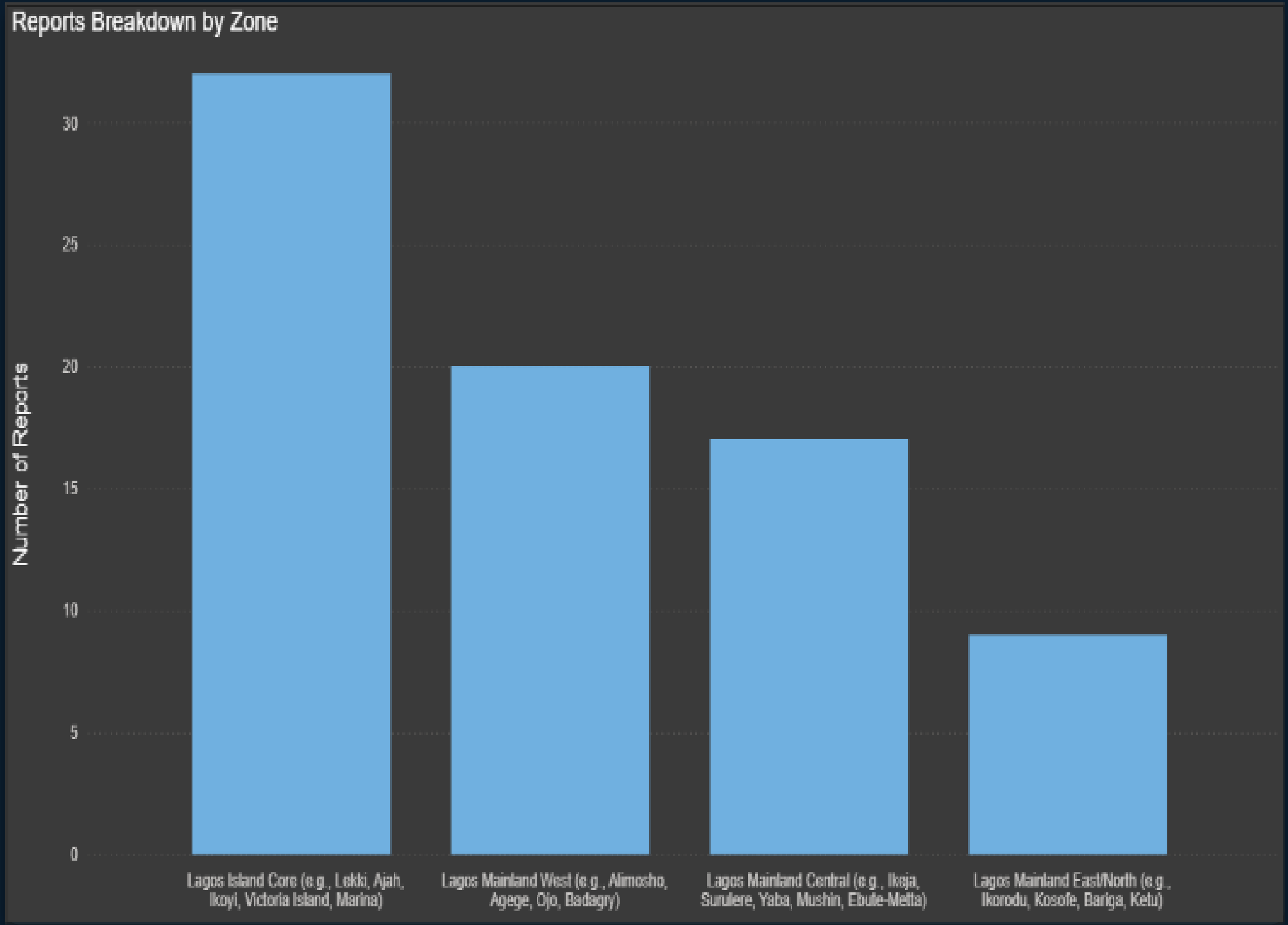
Lekki Phase 1 stands out as the single most-reported hotspot — more reports than the next three locations combined. This is where flood response, and the app's alerts, need to concentrate first.



03

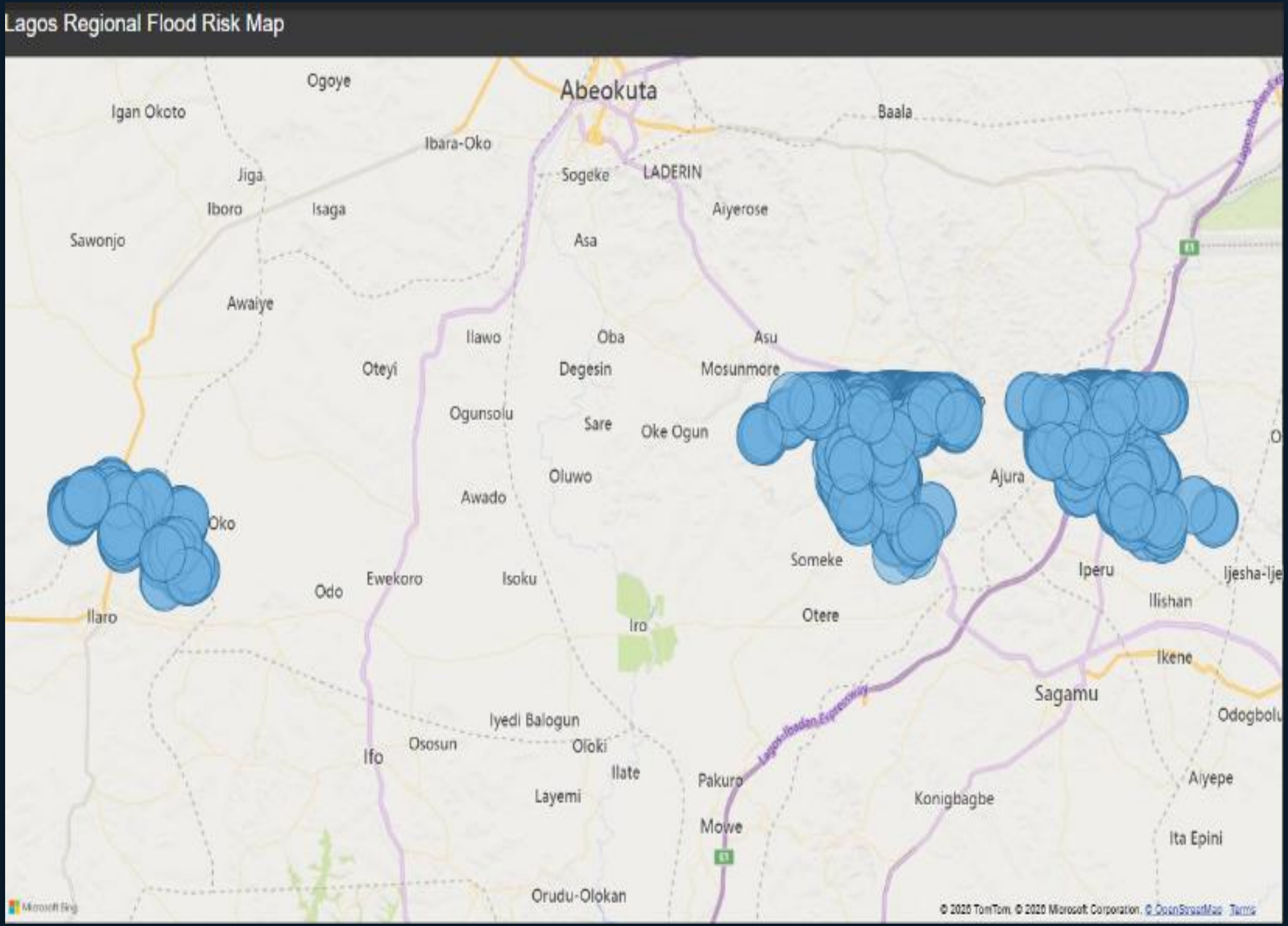
A Citywide Pattern

Zoom out from individual streets and the pattern holds citywide: Lagos Island Core absorbs the highest report volume, nearly double the other zones. Flooding isn't scattered randomly — it clusters where drainage and elevation make residents most vulnerable.



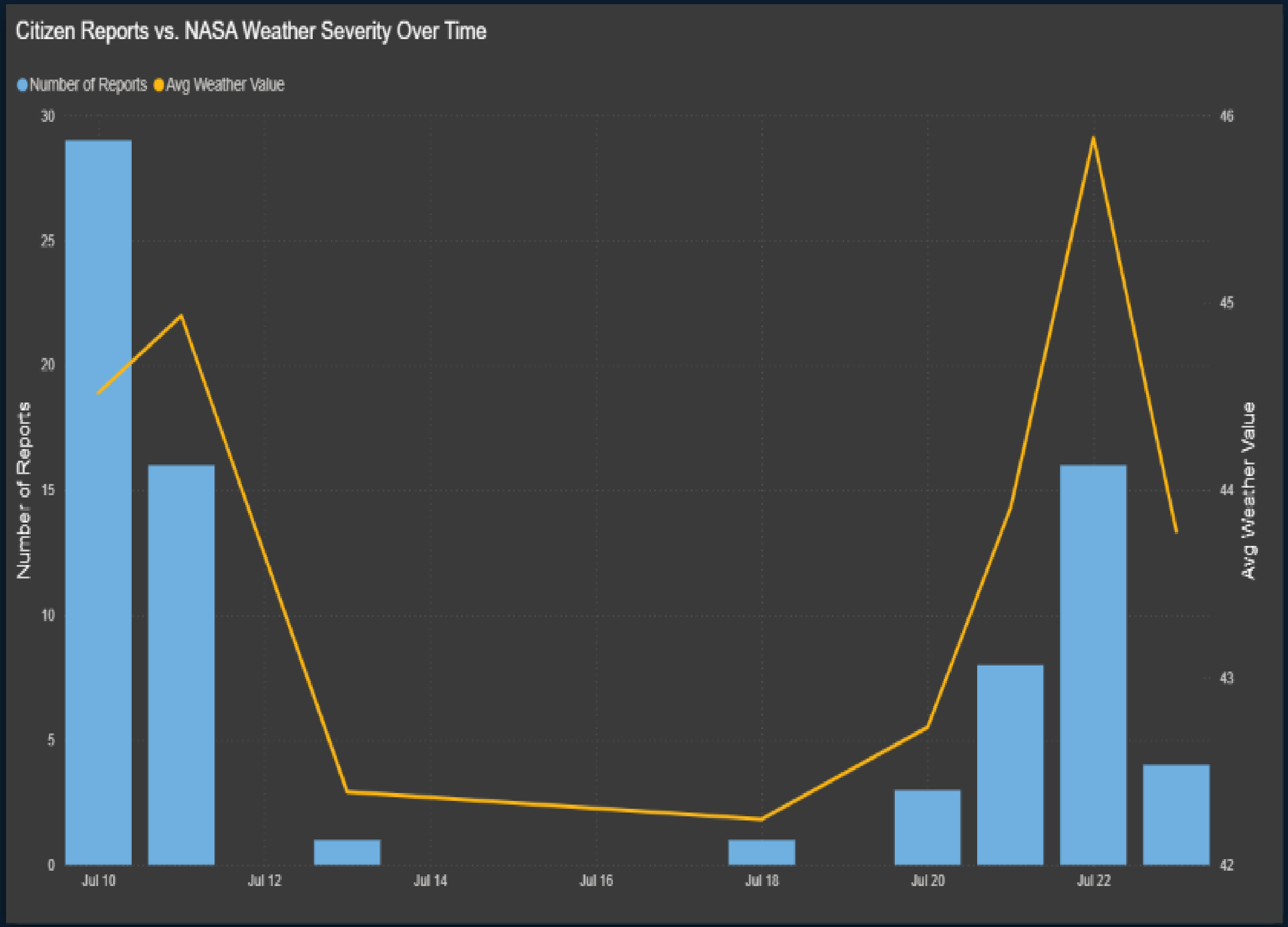
04 Mapping the Risk

Plotting every report on the map turns the pattern into geography: three distinct clusters emerge across Lagos's mainland and island zones — exactly the areas FloodWatch needs to prioritize for real-time monitoring.



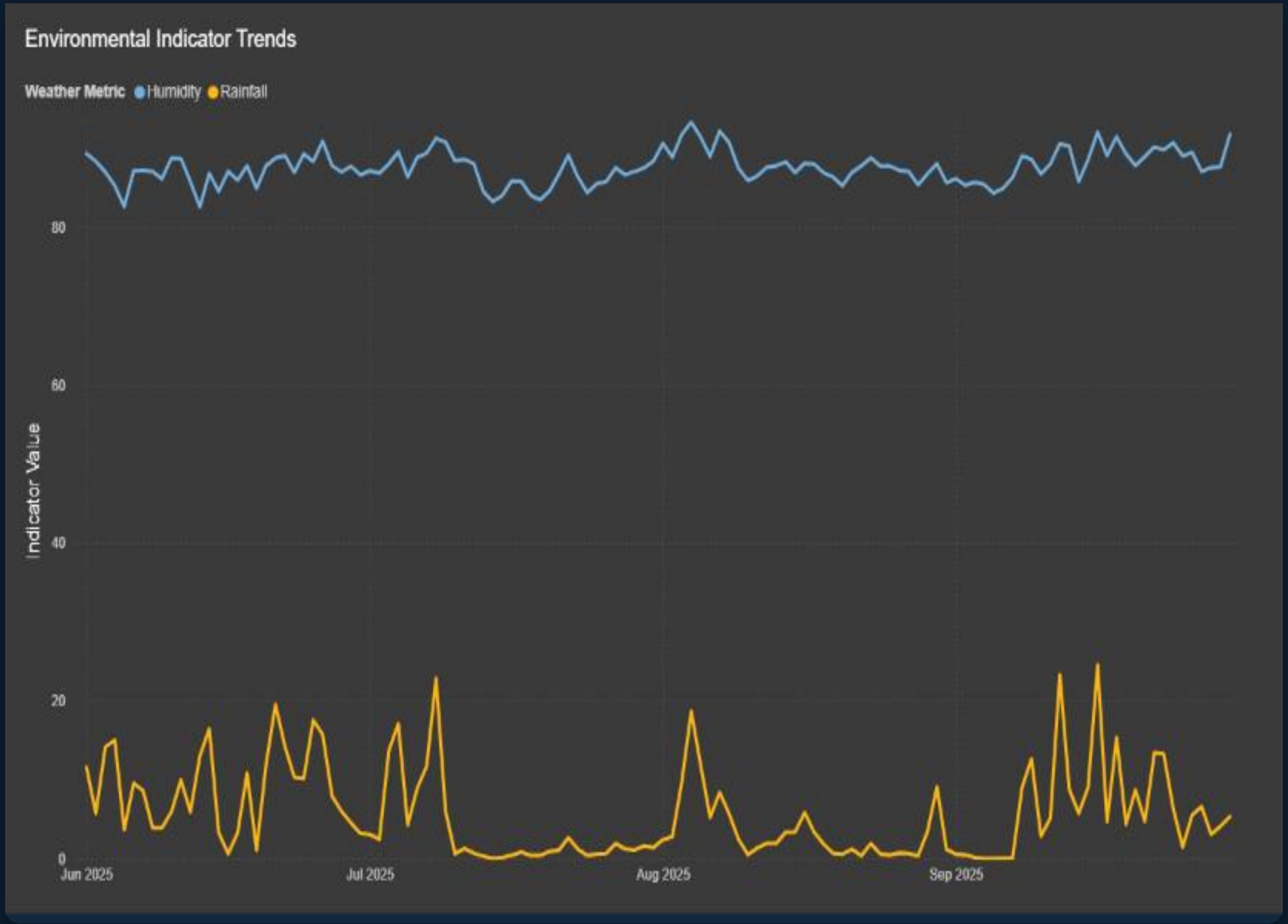
05 Reports Track the Rain

Citizen reports don't happen at random — they spike in near lock-step with NASA's weather severity readings. When the weather metric climbs, flood reports follow, confirming residents are reporting real conditions, not noise.



06 Rainfall Drives the Spikes

Humidity stays fairly constant across months, but rainfall is the volatile variable — and its spikes line up with the flood-reporting surges from the previous slide. Rainfall, not humidity, is the metric to design around.



07

How Residents Experience It

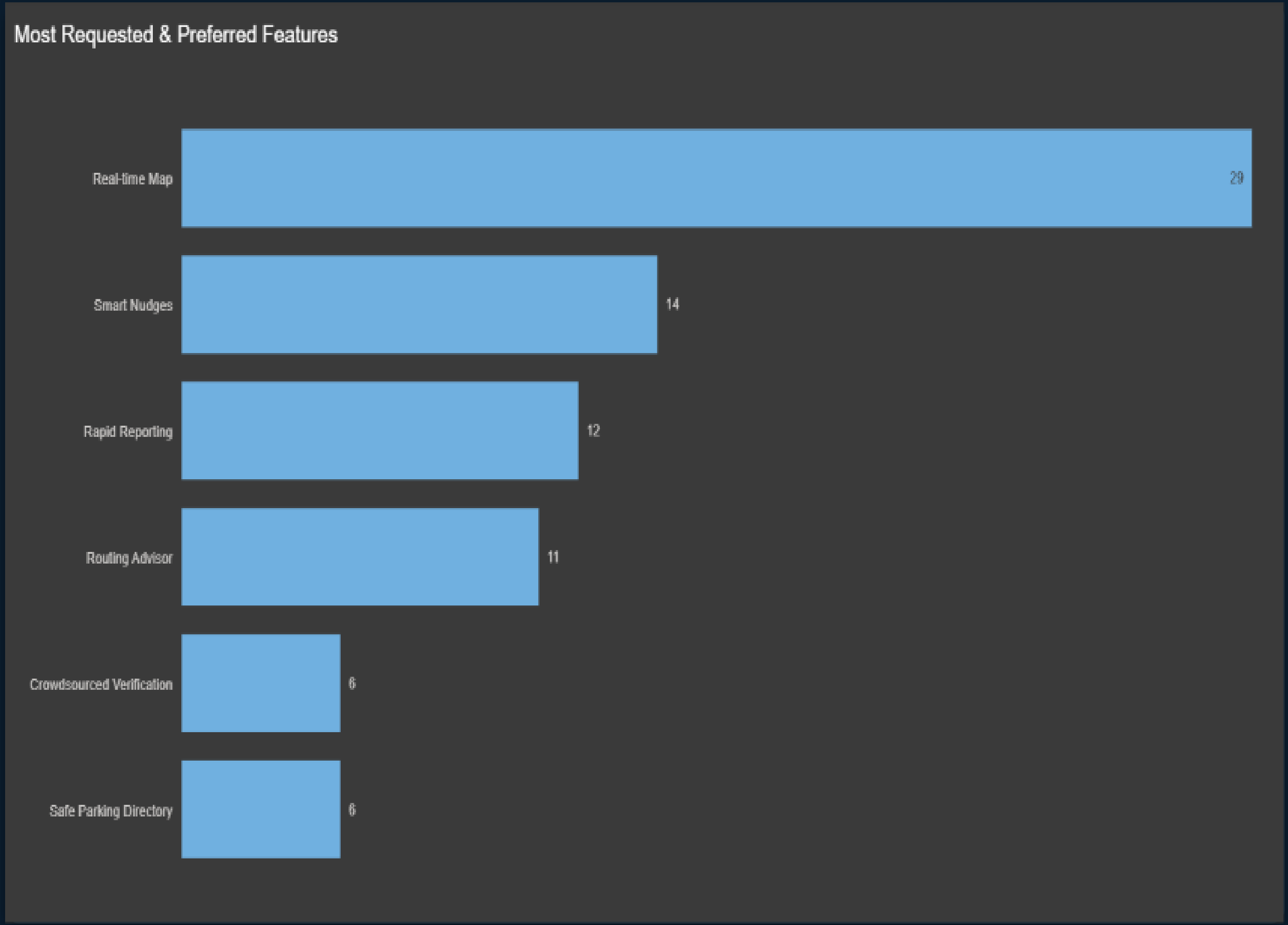
Ask residents directly and the picture sharpens: the largest group says flooding happens every single time it rains heavily — not only during rare, extreme storms. This is a frequent, everyday disruption.

08 Meet Our Community

These insights come from 37 residents surveyed directly — and 95% of them said they'd willingly report a flood again. A strong signal that the community wants to participate, if given the right tool.

09 What They Want From FloodWatch

When asked directly, a real-time map dominates every other request — more than double the next most-wanted feature, smart nudges. This should anchor the app's core experience.



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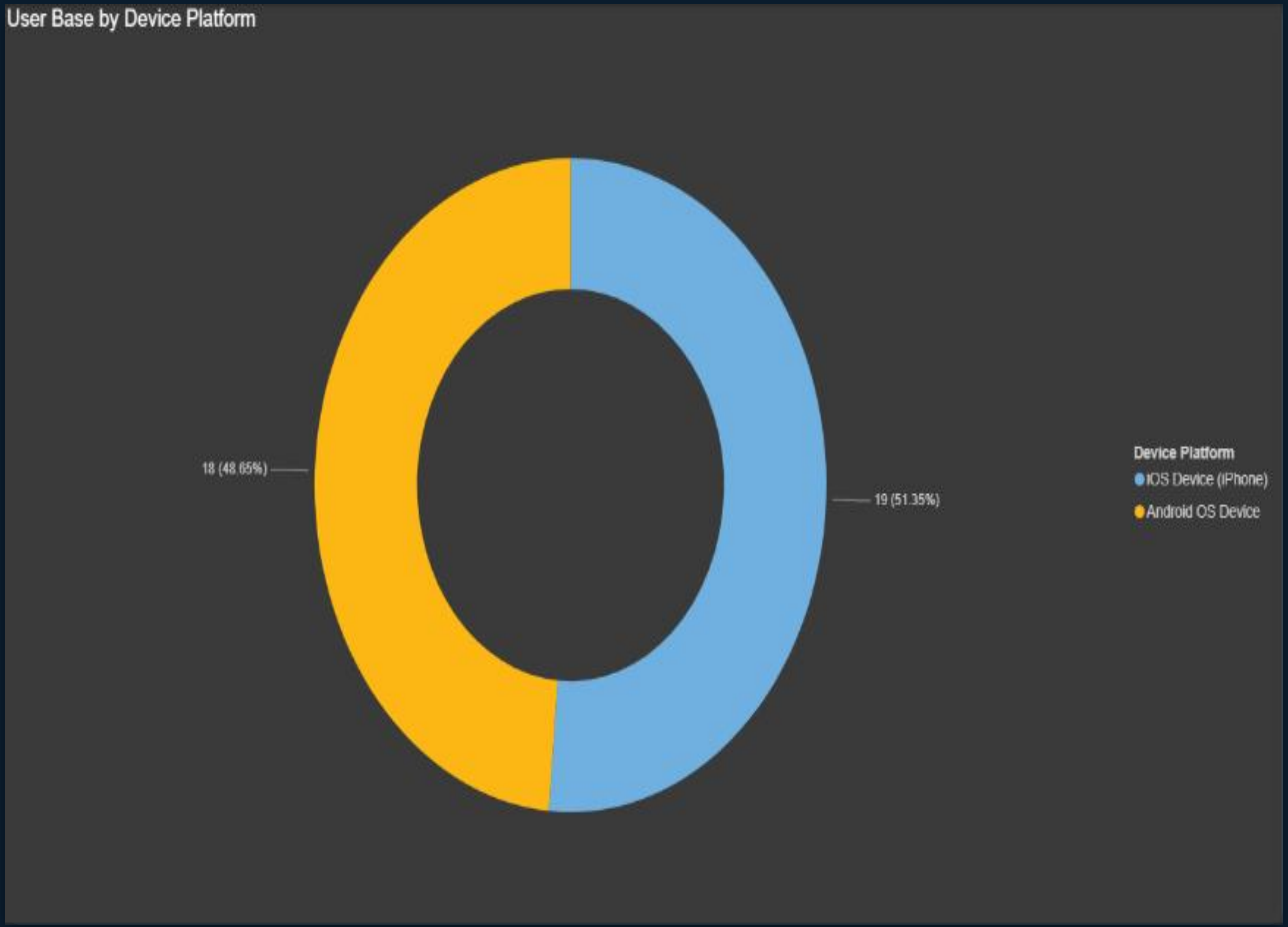
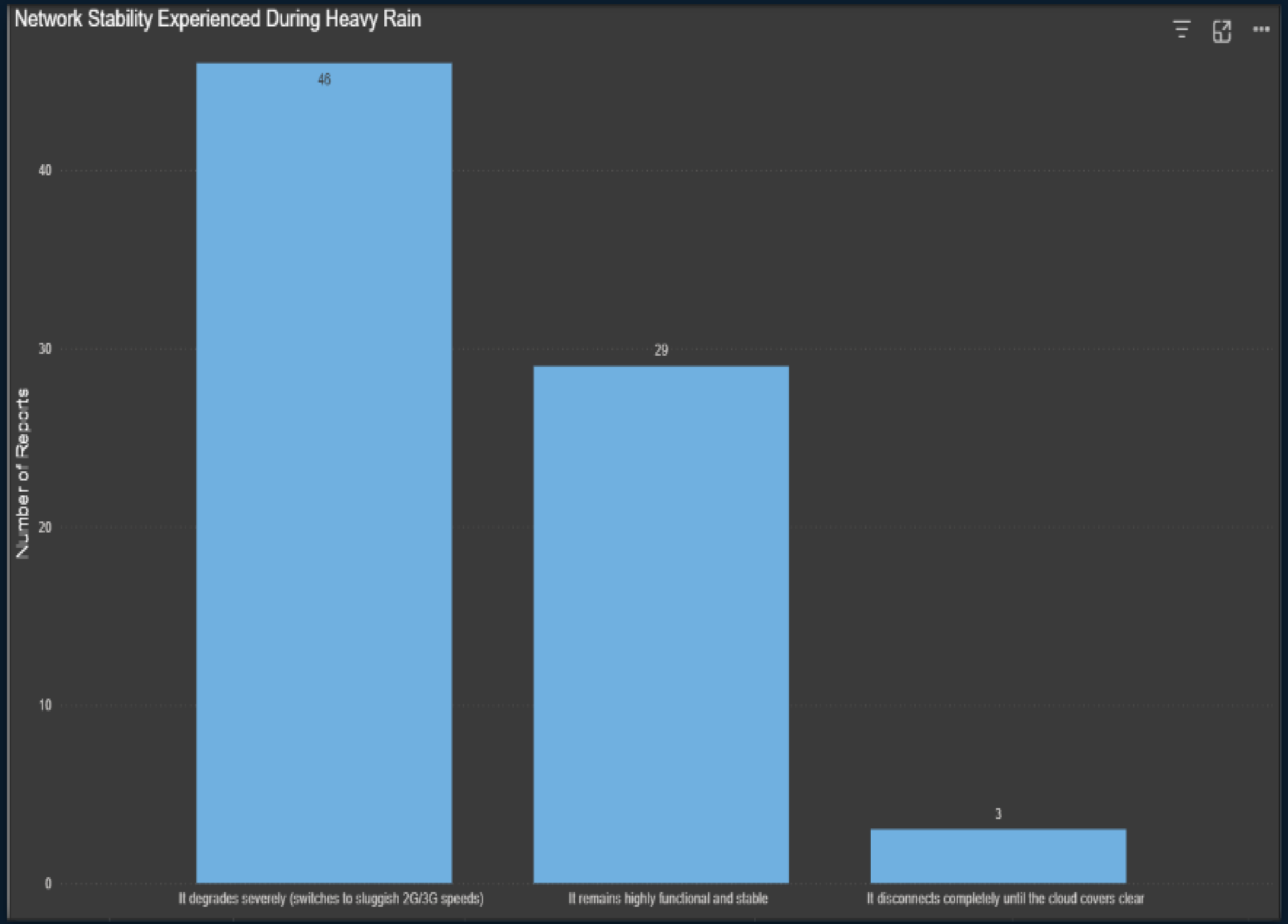
Built For Every Phone

The user base splits almost exactly down the middle between iOS and Android — 48.6% to 51.4%. Neither platform can be an afterthought; FloodWatch needs full feature parity on both.



11 The Connectivity Challenge

Here's the catch: during heavy rain, network service degrades severely for most users, and a small share loses connection entirely. Any real-time feature has to keep working even when the network doesn't.



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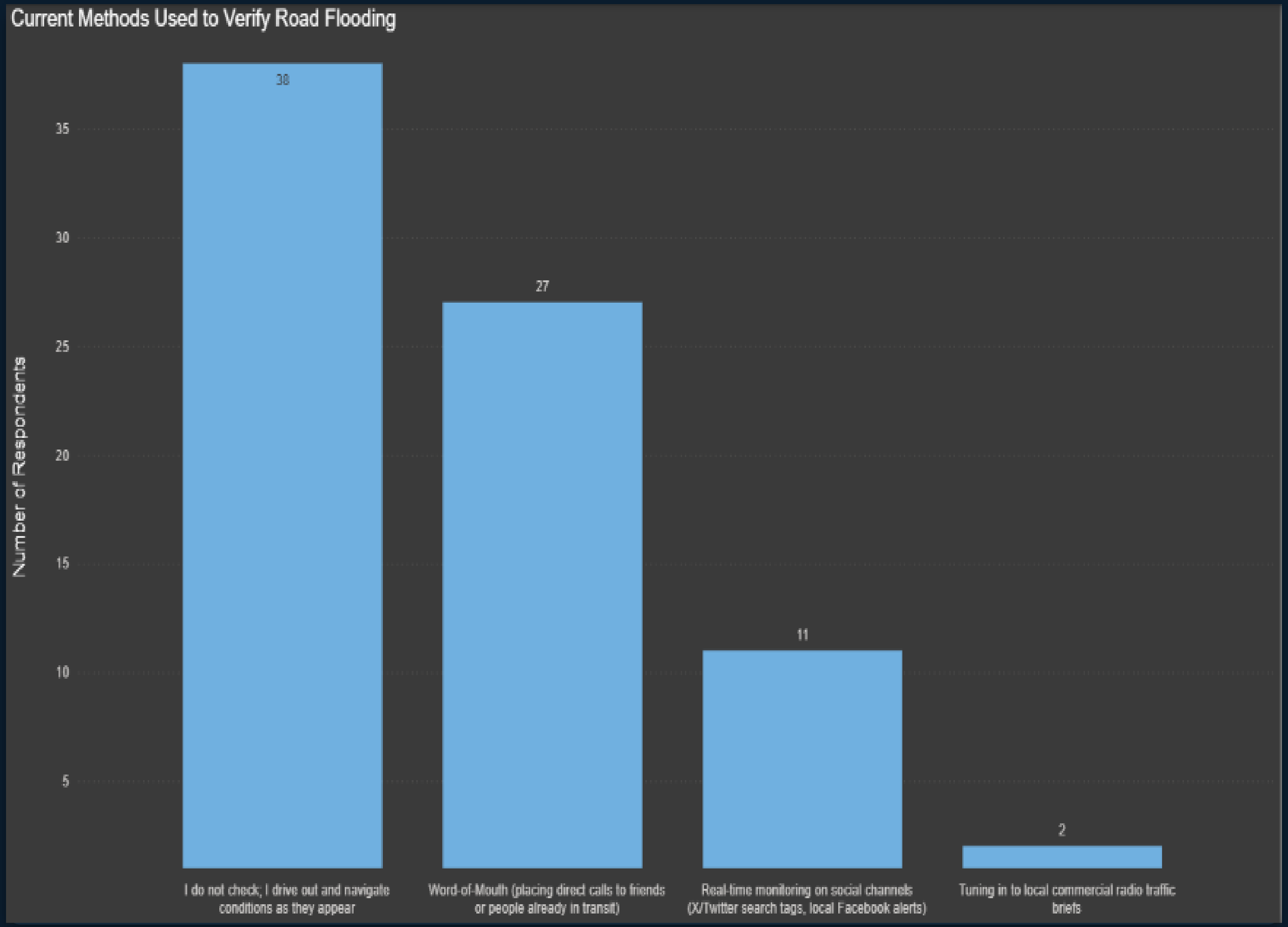
Why a Better Way Is Needed

Right now, most residents rely on the least reliable method available — driving out to check conditions themselves — followed by word-of-mouth. Almost no one uses real-time monitoring. This is exactly the gap FloodWatch closes.



From Data to Action

- 78 reports, 11 hotspots, one clear priority zone: Lekki Phase 1
- Rainfall — not humidity — is the leading indicator to design around
- Residents want a real-time map that keeps working when the network doesn't

Thank you.

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Thank you.